

Project 1 - Part 2

Student Information Management System (SIMS)

Programmer: Branson Zemaitis

Regression Testing Summary

All three required test cases (Alice Johnson, Michael Brown, Christopher Williams) were run against the compiled program and produced output matching the PDF's expected output exactly, covering both the original Part 1 functionality (Student ID/Name/GPA entry and summary) and the new Part 2 functionality (five course grades, average, highest, and lowest) in the same run. Screenshots of each run follow on the next pages.

Bugs Encountered and How They Were Corrected

No new bugs were encountered while implementing Part 2. During implementation, a structural mistake was caught and corrected before testing: the "Course Grades" report header was initially printed twice — once before the five grade prompts and once again in the final report — instead of only once, in the final report (matching the PDF's own expected output, which shows it only alongside the five "Course N :" lines). This was fixed by removing the duplicate header before the prompts.

Part 1's previously-fixed bug (stdout buffering causing jumbled prompts outside a real terminal, corrected with `fflush(stdout)` after each prompt) was re-verified to still be fixed — all five new grade prompts also call `fflush(stdout)` immediately after printing, for the same reason.

Answers to the Questions for the Customer

1. Should grades be stored as integers or decimal values?

Integers. Every worked example in the PDF uses only whole-number grades (95, 88, 91, 84, 100, 75, 82, 68, 79), so grades are entered, stored, and validated as whole numbers.

2. What grading scale is being used (0-100, percentage, points, etc.)?

0-100, matching every grade value shown in the PDF's worked examples.

3. Can grades contain fractional values such as 89.5?

No. All examples use whole-number grades, so fractional grades are not accepted at this stage; a grade is validated as a whole number between 0 and 100.

4. What should happen if an invalid grade is entered?

The program prints a clear error message identifying which course's grade was invalid and exits immediately, the same approach used for invalid Student ID/GPA input in Part 1. Re-prompting the user would require a loop, which remains out of scope this week.

5. Should the five courses have names, or only grades?

Only grades. The PDF's expected output labels them generically as "Course 1" through "Course 5" with no course-name field requested anywhere in the requirements.

6. Should the average be rounded to a specific number of decimal places?

Yes, two decimal places, matching the PDF's own example outputs (91.60, 79.00, 100.00).

7. What should happen if two courses share the highest grade?

The Highest Grade value is simply displayed once (e.g. 100). The Acceptance Criteria only requires identifying the highest grade value, not which course(s) earned it.

8. What should happen if two courses share the lowest grade?

Handled the same way as a tied highest grade — the Lowest Grade value is displayed once.

9. Should the report identify which course earned the highest and lowest grades?

No. This is not listed in the Acceptance Criteria, and identifying the specific matching course would effectively be a "searching" feature, which this week's spec explicitly places out of scope.

10. Will future versions support more than five courses?

Not addressed this week. The spec fixes the course count at five for every student enrolled this semester; supporting a variable number of courses would need to be requested in a future Client Change Request.

Test Case 1 — Alice Johnson

Expected: Average 91.60, Highest 100, Lowest 84.

Result: PASS — output matches PDF's expected output exactly.

```
terminal — sims

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Student Information Management System
Version 2.0

Programmer: Branson Zemaitis

Welcome to SIMS
-----

Enter Student ID: 1234567
Enter Student Name: Alice Johnson
Enter Current GPA: 3.84

Student Summary

Student ID : 1234567
Student Name : Alice Johnson
Current GPA : 3.84
Enter Course 1 Grade: 95
Enter Course 2 Grade: 88
Enter Course 3 Grade: 91
Enter Course 4 Grade: 84
Enter Course 5 Grade: 100
-----

Course Grades

Course 1 : 95
Course 2 : 88
Course 3 : 91
Course 4 : 84
Course 5 : 100

-----

Average Grade : 91.60
Highest Grade : 100
Lowest Grade : 84
```

Test Case 2 — Michael Brown

Expected: Average 79.00, Highest 91, Lowest 68.

Result: PASS — output matches PDF's expected output exactly.

```
terminal — sims

-----
Student Information Management System
Version 2.0

Programmer: Branson Zemaitis

Welcome to SIMS
-----

Enter Student ID: 1002003
Enter Student Name: Michael Brown
Enter Current GPA: 2.91

Student Summary

Student ID : 1002003
Student Name : Michael Brown
Current GPA : 2.91
Enter Course 1 Grade: 75
Enter Course 2 Grade: 82
Enter Course 3 Grade: 91
Enter Course 4 Grade: 68
Enter Course 5 Grade: 79
-----

Course Grades

Course 1 : 75
Course 2 : 82
Course 3 : 91
Course 4 : 68
Course 5 : 79

-----

Average Grade : 79.00
Highest Grade : 91
Lowest Grade : 68
```

Test Case 3 — Christopher Williams

Expected: Average 100.00, Highest 100, Lowest 100.

Result: PASS — output matches PDF's expected output exactly.

```
terminal — sims

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Student Information Management System
Version 2.0

Programmer: Branson Zemaitis

Welcome to SIMS
-----

Enter Student ID: 9876543
Enter Student Name: Christopher Williams
Enter Current GPA: 4.00

Student Summary

Student ID : 9876543
Student Name : Christopher Williams
Current GPA : 4.00
Enter Course 1 Grade: 100
Enter Course 2 Grade: 100
Enter Course 3 Grade: 100
Enter Course 4 Grade: 100
Enter Course 5 Grade: 100
-----

Course Grades

Course 1 : 100
Course 2 : 100
Course 3 : 100
Course 4 : 100
Course 5 : 100

-----

Average Grade : 100.00
Highest Grade : 100
Lowest Grade : 100
```